An Analysis of Spatial Clustering in MLB Batted Ball Distributions

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Introduction

Understanding how Major League Baseball (MLB) players hits are distributed across the diamond is central to evaluating defensive strategies. While traditional statistics summarize performance outcomes, they often overlook the spatial dimension of hitting.

This project investigates whether MLB hitters exhibit spatial clustering in their batted ball locations, and how these patterns of concentration vary among a selection of high-volume hitters. We are using data from the 2023 MLB regular season. High-volume hitters will be defined as any player with a minimum of 500 plate appearances and a minimum of 100 batted balls in play.

To implement this analysis, we will employ spatial clustering methods such as density-based clustering or kernel density estimation to identify regions of high batted ball concentration. Then we will assess statistical significance through a series of permutations, determining whether clustering patterns deviate from randomness.

Literature Background

The increasing availability of high-resolution tracking data has transformed the metrics and analysis of performance in Major League Baseball. Statcast, as described by the MLB is a "state-of-the-art tracking technology" that enables researches to investigate aspects of gameplay beyond the traditional box scores "in ways that were never possible in the past" (MLB 2026). Some of these granular aspects include swing mechanics, and ball flight. Statcast was originally implemented in 2015 consisting of a combination of cameras and radar systems, however in 2020, the MLB added the Hawk-Eye, a high-speed camera allowing for a replay system and various new features.

One substantial body of work examines batting performance and offensive prediction, leveraging Statcast data including "launch angle, launch velocity, and hit distance" (Bailey 2017). Using logistic regression to estimate the hit probabilities, these predictions are then aggregated to forecast future batting averages. Importantly, Bailey's analysis utilized Statcast data in combination with Player Empirical Comparison and Optimization Test Algorithm (PECOTA), a prediction system that considers previous seasons to predict batting averages. A linear regression was conducted on the relationship between the previous PECOTA and new Statcast data to create an even stronger metric for predicting batting averages.

Data and Methodology

To arrive at the 15 players of interest for this study, MLB players with at least 500 plate appearances (PAs) in the 2023 season were identified. 136 players met this criteria. Then batted ball dispersion data for each player was collected, and this data captures how frequently each player hits the ball to each field direction. The metrics of interest were pull percentage (pull%), opposite-field percentage (oppo%), and straight ahead percentage (cent%). Then each player was classified into one of four archetypes: balanced, center, opposite-field (oppo), or pull. If a player's pull%, oppo%, or cent% is larger than 40% they were classified into the associated archetype of the direction the tend to hit the most. If a player's pull%, oppo%, and cent% were all less than 40% this hitter would be classified as a balanced hitter.

75 of the players were classified as pull hitters, 51 as balanced hitters, 9 as center hitters, and 1 as an opposite field hitter. A stratified sampling approach was then used to ensure representation across all player archetypes. Based on proportional allocation from the full sample of 136 players, the initial values were approximately 8.27 pull hitters, 5.63 balanced hitters, 0.99 center hitters, and 0.11 opposite-field hitters given a sample size of 15 total players.

These values were then rounded to produce the final sample. Center and opposite-field hitters were both rounded to 1 ensuring that each archetype is represented in the sample. This rounding occurred because excluding the smaller groups would eliminate meaningful variation in hitting tendencies. Pull and balanced hitters were rounded down to 8 and 5 to maintain the total sample size of 15 while keeping the distribution as close as possible to the original proportions.

A final sample size of fifteen was chosen to balance representativeness of the overall 136 player population and to ensure the dataset remained manageable for spatial analysis.

The fifteen players chosen are as follows. Pull Hitters: Tyler Stephenson-R, Whit Merrifield-R, Jake Cronenworth-L, Seiya Suzuki-R, CJ Abrams-L, Josh Lowe-L, Josh Jung-R, Teoscar Hernández-R. Balanced hitters: Trent Grisham-L, Pete Alonso-R, Eugenio Suárez-R, TJ Friedl-L, J.T. Realmuto-R. Center Hitter: Yandy Díaz-R. Opposite-Field Hitter: Bo Bichette-R.

The side from which a player bats was not considered in constructing the sampling method. Nevertheless, the resulting sample which consisted of 5 left-handed batters and 10 right-handed batters approximately reflects the overall distribution of handedness in Major League Baseball of 30.3% (Grondin et al., 1999, as cited in Nachtigal & Barnes, 2019).